

# The Full Pressure Positivity Range at Every Integer Order

Research note prepared for Vladimir Reshetnikov with OpenAI

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## Abstract

For every integer  $m \geq 2$ , every even Taylor coefficient of integer Thue–Morse pressure at a degree strictly between  $2m$  and  $6m$  is positive. A weighted complex Green estimate, retaining the weight in a repaired inverse-branch residual, proves the whole range analytically for  $m \geq 112$ . Directed integer enclosures complete the remaining 110 orders, certifying all 12320 required coefficients. The finite calculation uses an explicit Eulerian spectral inverse, rigorously propagated error radii, and an independent Fourier residual verification. The upper endpoint is sharp for a universal statement: the degree-12 coefficient at  $m = 2$  is negative.

**Status.** This is unrefereed ordinary mathematics with rigorous finite arithmetic certificates, not a Lean formalization. No priority claim is made. The source package supplies the proof, reproducible programs, exact scalar certificates, pressure intervals, and per-order state hashes.

## 1 Statement and local transfer model

Put  $d = 2m$  and define

$$(V_a f)(x) = \sum_{2y \equiv x \pmod{1}} (\cos \pi y + a \sin \pi y)^d f(y).$$

The invariant Fourier space has modes  $1 - m, \dots, m - 1$ . At  $a = 0$ , its normalized positive fixed vector is

$$h_0(x) = \sin^d(\pi x) \sum_{n \in \mathbb{Z}} (\pi(x + n))^{-d}, \quad h_0(0) = 1.$$

The atomic eigenvalues are  $1, 2^{-1}, \dots, 2^{-(d-2)}$ ; the principal parts of the periodized poles give the corresponding eigenbasis. The local analytic eigenfunction  $h_a$ , normalized by  $h_a(0) = 1$ , and its eigenvalue  $\lambda(a)$  satisfy

$$\lambda(a) = 1 + a^d h_a(1/2), \quad p_m(t/\pi) = d \log \cos t + \log \lambda(\tan t). \quad (1)$$

These are the normalizations of the repository manuscript [1]. Write

$$H_{m,r} = [a^{2r}] h_a(1/2), \quad E_{m,r} = [t^{2m+2r}] p_m(t/\pi).$$

**Theorem 1.1.** *For every integer  $m \geq 2$  and every integer  $1 \leq r < 2m$ ,*

$$E_{m,r} > 0.$$

*Equivalently, every even pressure coefficient at degree  $N$  with  $2m < N < 6m$  is strictly positive.*

The coefficient at degree  $2m$  is the known cancellation coefficient and is not included. The preceding positive-triangle theorem gives  $H_{m,r}, E_{m,r} > 0$  for  $1 \leq r < m$  [2]; the first feedback boundary was treated separately in [3]. The analytic argument below covers the whole second-cluster interval for  $m \geq 112$ ; the exact finite proof covers  $2 \leq m \leq 111$ . The case  $m = 1$  is excluded because its pressure is identically zero.

*Editorial note (ProveIt, 2026-10-01).* Theorem 1.1 contains the pressure statements of the earlier packages of this series: `../Thue_Morse_Integer_Pressure_First_Positive/` ( $r = 1$ ), `../Thue_Morse_Integer_Pressure_Higher_Positive/` ( $r = 2$ , and  $m \geq 3r$ ), `../Thue_Morse_Integer_Pressure_Positive_Triangle/` ( $r < m$ ; cited here as [2]), `../Thue_Morse_Integer_Pressure_Feedback_Boundary/` ( $r = m$ ; re-proved here), and `../Thue_Morse_Integer_Pressure_Beyond_Boundary/` and `../Thue_Morse_Integer_Pressure_Linear_Region/` ( $r = m + s$  with  $s < m$ ). Their response statements ( $H_{m,r} > 0$  for  $r \leq m$  and in the two regions beyond, and the diagonal and fixed-offset asymptotics) are not contained in it.

Set  $\text{ev } f = f(1/2)$ ,  $R_m = \text{ev } h_0$ ,  $Pf = f(0)h_0$ , and  $Q = I - P$ . On the subspace of functions vanishing at zero put

$$\mathcal{B}_a = QV_aQ, \quad q_a = QV_a h_0, \quad F(a) = R_m + \text{ev}(I - \mathcal{B}_a)^{-1}q_a, \quad G(a) = \text{ev}(I - \mathcal{B}_a)^{-2}q_a.$$

These initially define analytic germs. Reflection makes  $F, G$  even, and  $G(0) = 0$ . If  $\delta = \lambda - 1$ , the projected eigenvalue equation gives

$$\delta = a^d \{R_m + \text{ev}(I - \mathcal{B}_a + \delta)^{-1}q_a\}.$$

Expanding once in  $\delta$  yields the exact formal identities

$$\begin{aligned} \delta &= a^d F - a^{2d} FG + O(a^{3d}), \\ \log \lambda &= a^d F - a^{2d} K + O(a^{3d}), \quad K = FG + F^2/2. \end{aligned} \tag{2}$$

Thus all pressure coefficients below degree  $3d$  are determined by

$$d \log \cos t + \tan^d t F(\tan t) - \tan^{2d} t K(\tan t). \tag{3}$$

In particular  $[a^q]F > 0$  for even  $0 \leq q < d$ , by the positive triangle, and  $[a^d]F = H_{m,m}$ .

## 2 A uniform bound for the quadratic cluster

Define

$$c = 2/\pi, \quad b_j = \tan(\pi/2^{j+1}), \quad B(a) = \prod_{j \geq 1} (1 + ab_j), \quad g(a) = caB(a).$$

The product  $B$  is entire. We use the dyadic tangent comparison proved in [2]: for odd  $n \geq 1$ , the decreasing rearrangement of

$$x_j(n) = |\tan(n\pi/2^{j+1})|/n$$

is weakly log-majorized by  $(b_j)$ . It follows that

$$[a^q] \prod_j (1 + ax_j(n))^d \leq M_q, \quad M_q = [a^q]B(a)^d. \tag{4}$$

For finite sequences, the implication follows because elementary symmetric functions of exponentials are symmetric increasing convex functions of the log coordinates; finite-prefix limits give the summable-sequence version. The earlier report contains the dyadic sine-product proof of the comparison itself and is included with the present sources.

**Lemma 2.1.** *For even  $N$  with  $2d \leq N < 3d$ ,*

$$\left| [t^N] \tan^{2d} t K(\tan t) \right| \leq C_K(d) [t^N] g(\tan t)^{2d}, \quad C_K(d) = \frac{25}{2} \log_2 d + \frac{385}{8}.$$

*Proof.* The constant coefficient is separate:  $G(0) = 0$  and  $F(0) = 2c^d \sum_{n \geq 1 \text{ odd}} n^{-d} < (5/2)c^d$ . In the following argument take  $1 \leq q \leq d-2$ . For a coefficient of degree  $q < d$ , every positive-degree insertion in  $V_a$  vanishes at zero, so the  $Q$  projections in the insertion expansion can be omitted. Unfolding the periodization of  $h_0$  gives

$$[a^q]F = \sum_{w \in \mathbb{Z} + 1/2} (\pi w)^{-d} [a^q] \prod_{j \geq 1} (1 + a \tan(\pi w/2^j))^d.$$

For  $G$ , each monomial has the additional weight equal to its deepest occupied factor index. Indeed, if  $F_L = \text{ev } V_a^L h_0$ , then coefficientwise below degree  $d$ ,

$$G = \sum_{L \geq 0} (F - F_L).$$

A monomial first appearing at depth  $J$  is counted exactly  $J$  times.

For a nonnegative sequence  $x$ , mark one chosen unit beyond depth  $J$ . Its degree- $q$  contribution is at most

$$d \left( \sum_{j > J} x_j \right) M_{q-1}.$$

There are  $d$  labeled copies of  $b_1 = 1$ , so adding one unused such copy gives

$$M_q \geq \frac{d - q + 1}{q} M_{q-1}.$$

For  $q \leq d-2$ , the marked tail is consequently at most  $\min(1, d^2 \sum_{j > J} x_j) M_q$ . For  $J \geq \lceil \log_2 n \rceil$ ,

$$\sum_{j > J} x_j(n) \leq \pi 2^{-J}.$$

Summing the depth indicators bounds the depth-weighted coefficient by

$$(\log_2 n + 2 \log_2 d + 6) M_q.$$

Pair  $w = \pm n/2$  and use

$$\sum_{n \geq 1 \text{ odd}} n^{q-d} \leq \pi^2/8 < 5/4, \quad \sum_{n \geq 1 \text{ odd}} (\log_2 n) n^{q-d} < 3/2.$$

The latter follows by comparison with  $\sum_{n \geq 2} (\log_2 n)/n^2$  and  $\log 2 > 2/3$ . We obtain

$$|[a^q]F| \leq \frac{5}{2} c^d M_q, \quad |[a^q]G| \leq (5 \log_2 d + 18) c^d M_q, \quad 0 \leq q \leq d-2.$$

These absolute bounds also justify the sums and interchanges above. Convolution bounds  $FG$  by  $((25/2) \log_2 d + 45) c^{2d} B^{2d}$  and  $F^2/2$  by  $(25/8) c^{2d} B^{2d}$  in the required coefficients. Nonnegative tangent coefficients complete the proof.  $\square$

### 3 A compact bound for signed branch products

Let

$$\mathcal{H}_a(x) = \prod_{j \geq 1} (\cos(\pi x/2^j) + a \sin(\pi x/2^j)).$$

This product converges locally uniformly in  $(a, x) \in \mathbb{C}^2$ . It satisfies

$$\mathcal{H}_a(x) = W_a(x/2) \mathcal{H}_a(x/2), \quad W_a(x) = \cos \pi x + a \sin \pi x, \quad \mathcal{H}_a(\pm 1/2) = cB(\pm a). \quad (5)$$

Fix  $R = 2/5$ , put  $h(t) = \mathcal{H}_R(t)$ , and write  $w = 49/100$ .



On  $[0, 1/4]$  the first term exceeds  $3/2$ , while  $\ell'(y) < 0$  and  $\ell'(x) < 13/10$ . On  $[1/4, 1/2]$  it is at least  $3\pi/14 > 3/5$ , and both relevant derivatives are below  $2/5$ . Hence  $b_-$  increases to  $b_-(1/2) = 1$ ; its value at  $2/5$  is certified below  $91/100$ . The function  $p$  decreases and its value at  $2/5$  is below  $14/25$ . The logarithmic derivative of  $b_+$  is negative before  $s_x = Rc_x$  and positive afterwards: bounds are  $-\pi/(2R) + 3/4 < 0$  and  $7\pi/6 - (3/2)(13/10) > 0$ . Its endpoint values are  $RL < 1/2$  and  $3/7$ , proving the remaining statement.

**Theorem 4.1.** *For even  $d \geq 128$  and  $|a| \leq R$ , the operator  $T_a = QV_a$  is a strict contraction in an explicit equivalent  $C^1$  norm on*

$$X = \{f \in C^1(\mathbb{R}/\mathbb{Z}) : f(0) = 0\}.$$

*Its norm is below  $4/5$ . Consequently the frozen response has no pole in this disk, and for  $s \in X$ ,*

$$|\text{ev}(I - T_a)^{-1}s| \leq 5L^d \|s'\|_\infty. \quad (10)$$

*Proof.* Let  $t(x) = \text{dist}(x, \mathbb{Z})$  and define

$$N_0(f) = \sup_{x \notin \mathbb{Z}} \frac{|f(x)|}{t(x)h(t(x))^d}, \quad N_1(f) = \sup_x \frac{|f'(x)|}{h(t(x))^d}.$$

These are finite, and every positive weighted sum  $N_0 + \eta N_1$  is equivalent to the  $C^1$  norm on  $X$ , for fixed  $d$ . For nonnegative  $A_0, A_1$ , the common-phase expression

$$A_0|c_x + as_x|^d + A_1|s_x - ac_x|^d$$

attains its maximum on  $|a| \leq R$  at  $a = R$  or  $a = -R$ . On the boundary it is a sum of convex functions of  $\cos \arg a$ , because  $d/2 \geq 1$ . For fixed  $\Re a$ , moving to the boundary increases both squared moduli. This observation is essential: separate branchwise maxima would lose the estimate.

By reflection it suffices to consider  $0 \leq x \leq 1/2$ . Also

$$T_a f = V_a f - a^d f(1/2) h_0, \quad \|h_0\|_\infty \leq 1, \quad \|h'_0\|_\infty \leq \pi d.$$

The first bound follows from  $d \geq 2$ ,  $|\sin(\pi x)/(\pi(x+n))| \leq 1$ , and the identity  $\sum_n (\sin(\pi x)/(\pi(x+n)))^2 = 1$ . The second follows from the trigonometric Bernstein inequality. The identity  $h(x) = (c_x + Rs_x)h(x/2)$  cancels the principal branch weight.

For  $x \geq 1/10$ , the two branch bounds in  $N_0$  are, at the two extreme phases,

$$\frac{1}{2} + \frac{1-x}{2x} b_+(x)^d, \quad \frac{1}{2} p(x)^d + \frac{1-x}{2x} b_-(x)^d.$$

The projection costs at most  $2^{-d}/(2x)$ . The resulting bounds are

$$\begin{cases} 1/2 + (9/2)(91/100)^d + 5 \cdot 2^{-d}, & 1/10 \leq x \leq 2/5, \\ 3/4 + (1/2)(14/25)^d + 2 \cdot 2^{-d}, & 2/5 \leq x \leq 1/2. \end{cases}$$

Near zero retain the rank-one cancellation. Put

$$u_a(x) = W_a((x+1)/2)^d f((x+1)/2).$$

The secondary projected term is  $u_a(x) - u_a(0) + a^d f(1/2)(1 - h_0(x))$ . On  $0 \leq x \leq 1/10$ ,

$$|W_a((x+1)/2)| < 39/70, \quad |\partial_x W_a((x+1)/2)| \leq 7\pi/10.$$

Since  $(39/70)\Lambda = 39/50 < 4/5$ , the mean-value bound gives

$$\frac{|u_a(x) - u_a(0)|}{x} \leq \frac{11d}{4} (4/5)^d N_0(f) + \frac{1}{2} (4/5)^d N_1(f).$$

The  $h_0$  correction costs at most  $(11d/7)2^{-d}N_0(f)$ , and the principal branch costs  $N_0(f)/2$ . Combining the intervals yields

$$N_0(T_a f) \leq \alpha_d N_0(f) + \varepsilon_d N_1(f), \quad \begin{cases} \alpha_d = \frac{3}{4} + (5d+8)(4/5)^d + \frac{9}{2}(91/100)^d, \\ \varepsilon_d = \frac{1}{2}(4/5)^d. \end{cases} \quad (11)$$

For  $N_1$ , derivatives landing on  $f$  cost at most  $\beta_d N_1(f)$ , with  $\beta_d = 1/2 + (1/2)(91/100)^d$ , by the same common-phase estimate. For derivatives landing on a weight, each individual normalized branch base is at most one, so

$$\frac{|W_a(y)|^{d-1} h(t(y))^d}{h(t(x))^d} \leq \Lambda.$$

The two weight derivatives cost at most  $(77/25)dN_0(f)$ , and the projection derivative at most  $(11/7)d2^{-d}N_0(f)$ . Therefore

$$N_1(T_a f) \leq 5dN_0(f) + \beta_d N_1(f). \quad (12)$$

Set  $\eta_d = (9/10)^d/d$  and  $\|f\|_d = N_0(f) + \eta_d N_1(f)$ . The two norm rows are bounded by

$$\alpha_d + 5(9/10)^d, \quad \beta_d + \frac{d}{2}(8/9)^d.$$

Exact rational checks at  $d = 128$ , and the decreasing successive ratios thereafter, put both below  $4/5$ . Thus the Neumann inverse has norm at most five. Since  $h \geq 1$ ,  $\|s\|_d \leq 2\|s'\|_\infty$ ; evaluation costs  $L^d/2$ . This proves (10).

The operator family is polynomial in  $a$  on this fixed Banach space. The strict uniform contraction extends its analytic inverse to a neighborhood of the closed disk. Its restriction to the invariant Fourier subspace agrees with the original frozen germ by uniqueness, proving the pole statement.  $\square$

## 5 The weighted nearest-pair residual

Put  $g = RL$ ; here  $g$  denotes this positive number, whereas  $g(a) = caB(a)$  remains the scalar generating function. The exact constants below give  $R < g < w$ .

**Theorem 5.1.** *For even  $d \geq 128$  and  $|a| \leq 2/5$ ,*

$$\left| F(a) - c^d(B(a)^d + B(-a)^d) \right| \leq 31000d^2(wL)^d < 26000d^2(7/10)^d. \quad (13)$$

*Proof.* On  $[0, 1]$  put  $\phi(x) = \mathcal{H}_a(x)^d + \mathcal{H}_a(x-1)^d$ . The product identity and even  $d$  give

$$V_a \phi - \phi = \mathcal{H}_a(x-2)^d + \mathcal{H}_a(x+1)^d =: \mathcal{R}_a(x). \quad (14)$$

On  $|x| \leq 2$ ,  $|a| \leq R$ , one has  $|\mathcal{H}'_a(x)| < 120$ . Indeed a product of any subset of factors is at most  $e^{R\pi|x|} < e^3 < 27$ ; summing the differentiated-factor bounds  $\pi(1+R)2^{-j}$  proves this estimate.

The endpoint values of  $\phi$  differ from one by at most  $g^d$ . Its common main endpoint derivative is  $d\pi a$ ; each excess derivative is at most  $120dg^{d-1} < 300dg^d$ . Let  $C_a$  be the cubic Hermite interpolant matching the two excess values and two excess derivatives. Then  $\psi = \phi - C_a$  is periodic  $C^1$ ,  $\psi(0) = 1$ , both endpoint derivatives equal  $d\pi a$ , and

$$\|C_a\|_\infty, \|C'_a\|_\infty \leq 610dg^d.$$

This specifies the Hermite convention used in every bound.

Set  $f_F = h_0 + (I - T_a)^{-1}QV_a h_0$  and  $e = f_F - \psi$ . The exact equation is

$$(I - T_a)e = s, \quad s = \mathcal{R}_a - (V_a C_a - C_a) - a^d \psi(1/2)h_0.$$

Here  $e, s \in X$  and  $s(0) = 0$ . Periodicity follows because the first two terms equal  $V_a\psi - \psi$ .

Retain the Green weight in the source. Lemma 3.1 and  $h \geq 1$  give  $N_1(\mathcal{R}_a) < 500dw^d$ . Each individual branch in the Green proof satisfies

$$|W_a(y)| \frac{h(t(y))}{h(t(x))} \leq 1.$$

Since  $h \geq 1$ , both  $|W_a(y)|^d/h(t(x))^d$  and  $|W_a(y)|^{d-1}/h(t(x))^d$  are at most one. Thus the unweighted Hermite repair incurs no exponential transfer loss. With  $J = \sqrt{1+R^2} < 27/25$  and  $\pi J < 17/5$ ,

$$N_1(V_a C_a) \leq \pi J d \|C_a\|_\infty + \|C'_a\|_\infty < 2700d^2 w^d, \quad N_1(C_a) \leq 610dw^d.$$

Also  $|\psi(1/2)| \leq 2L^d + 610dg^d$ . The rank-one source derivative is bounded by  $7dw^d + 2000d^2w^d$ . Adding the four terms gives

$$N_1(s) \leq 6000d^2w^d.$$

For  $t \leq 1/4$ , the function  $h$  increases, by strict concavity of  $\log h$  and  $\ell'(1/4) > 0$ . Integrating from the nearest integer gives the same bound for  $N_0(s)$  in that region. For  $t \geq 1/4$ , the direct weighted amplitudes of the four source terms are bounded respectively by

$$2w^d, \quad 1220dw^d, \quad 610dw^d, \quad (2 + 610d)w^d.$$

Division by  $t \geq 1/4$  gives at most  $10000dw^d \leq 6000d^2w^d$ . Hence  $\|s\|_d \leq 12000d^2w^d$ . The inverse has norm at most five, and evaluation costs  $L^d/2$ . Adding  $|C_a(1/2)|$  proves the first bound in (13).

The rational scalar check gives  $L < 243/200$ , so  $wL < 3/5$ . Since  $(31/26)(6/7)^{128} < 1$ , the second bound in (13) follows for every  $d \geq 128$ . No finite Fourier assumption is imposed on the repaired functions: the  $C^1$  Green theorem justifies their use.  $\square$

## 6 Log-concavity and a coefficient lower bound

**Lemma 6.1.** *If a positive log-concave integer law  $(p_j)_{j \geq 0}$  has integer mean  $s$  and finite exponential moment, then*

$$p_s \geq \frac{1}{\sqrt{2\pi e(\text{Var } X + 1/12)}}.$$

*Proof.* The piecewise-linear interpolation of  $\log p_j$  is concave. Jensen gives  $\log p_s \geq \sum_j p_j \log p_j = -H(X)$ . If  $U$  is independent uniform on  $[-1/2, 1/2]$ , its jittered density satisfies  $h(X+U) = H(X)$  and has variance  $\text{Var } X + 1/12$ . The nonnegativity of relative entropy against the Gaussian with this mean and variance gives  $H(X) \leq \frac{1}{2} \log(2\pi e(\text{Var } X + 1/12))$ . This inequality also follows directly by integrating  $f \log(f/g) \geq f - g$  over the support of  $f$ . Exponential moments justify all integrals and Jensen's expectation.  $\square$

**Lemma 6.2.** *The coefficients of*

$$T(x) = \left( \frac{\tan \sqrt{x}}{\sqrt{x}} \right)^2$$

*are positive and log-concave. For integers  $k \geq 1$  and  $0 \leq s \leq k/2$ ,*

$$[x^s]T(x)^k \geq \frac{4^s}{5\sqrt{k}}. \tag{15}$$

*Proof.* Write  $R_j = 2c^{2j}Z(2j)$ , where  $Z(u) = \sum_{n \geq 1 \text{ odd}} n^{-u}$ . Differentiating tangent gives  $[x^n]T = (2n+3)R_{n+2}$ . For  $u \geq 4$ ,

$$Z(u) - 1 \leq 3^{-u} + \frac{1}{2} \int_3^\infty v^{-u} dv < 2 \cdot 3^{-u}.$$

For  $n \geq 1$ , put  $z = 3^{-2n-2}$ . Then

$$\frac{R_{n+1}R_{n+3}}{R_{n+2}^2} \leq (1+2z)^2 \leq 1+5z < 1 + \frac{4}{(2n+1)(2n+5)} = \frac{(2n+3)^2}{(2n+1)(2n+5)}.$$

The strict inequality holds at  $n = 1$  and propagates because  $(2n+3)(2n+7) < 9(2n+1)(2n+5)$ . This proves coefficient log-concavity. Convolution preserves it: the Toeplitz matrix of a nonnegative log-concave sequence without internal zeros has nonnegative two-by-two minors, and matrix multiplication plus the two-by-two Cauchy–Binet identity proves closure. Finite coefficient limits handle infinite sequences.

For  $s > 0$ , choose the positive tilt  $x = t^2$  for which the law with PGF  $T(xz)^k/T(x)^k$  has mean  $s$ . Its mean and variance per factor are

$$\mu(t) = \frac{t}{\sin t \cos t} - 1, \quad v(t) = \frac{t}{\sin 2t} - \frac{2t^2 \cos 2t}{\sin^2 2t}.$$

Since  $\mu(t) = s/k \leq 1/2$  and  $\mu(\pi/4) = \pi/2 - 1 > 1/2$ , the tilt lies below  $\pi/4$ . Hence  $v(t) \leq t/\sin 2t = (\mu(t) + 1)/2 \leq 3/4$ . Lemma 6.1, with  $\pi < 22/7$  and  $e < 3$ , gives mass at the integer mean at least  $1/(5\sqrt{k})$ .

Put

$$\Psi(\gamma) = \inf_{0 < t < \pi/2} \{2 \log \tan t - \gamma \log t\}.$$

This is concave,  $\Psi(2) = 0$ , and  $\Psi(3) > \log 2$ . For the last fact, the positive Taylor lower bound  $\tan t \geq t + t^3/3 + 2t^5/15$  and the exact inequality

$$(15 + 5t^2 + 2t^4)^2 - 450t > 0 \quad (t > 0)$$

imply  $\tan^2 t > 2t^3$  on  $0 < t < \pi/2$ . The polynomial inequality is verified by the supplied exact Sturm sequence, which has no positive root. Concavity gives  $\Psi(\gamma) \geq (\gamma - 2) \log 2$  for  $2 \leq \gamma \leq 3$ . The tilted normalization at  $\gamma = 2 + 2s/k$  is therefore at least  $4^s$ . This proves (15);  $s = 0$  is immediate.  $\square$

Let  $a_* > 0$  be defined by  $a_* B'(a_*)/B(a_*) = 1$ , and  $C_* = B(a_*)/a_*$ . Positive coefficients and summable  $b_j$  give uniqueness. The rational half-angle checks show

$$0.822 < a_* < 0.823, \quad C_* > 101/25, \quad cC_* > \alpha := 707/275, \quad \arctan(2/5) > 19/50. \quad (16)$$

The law with PGF  $B(a_* z)^d/B(a_*)^d$  is an infinite Poisson-binomial law, with mean  $d$  and variance below  $d$ . Its coefficients are log-concave by finite Bernoulli convolutions and limits. Lemma 6.1 gives

$$[a^d]B(a)^d \geq C_*^d/(5\sqrt{d}).$$

Cauchy's estimate in (13) therefore yields

$$H_{m,m} = [a^d]F \geq \frac{2\alpha^d}{5\sqrt{d}} - 26000d^2(7/4)^d > \frac{\alpha^d}{4\sqrt{d}}, \quad d \geq 128.$$

The final comparison follows from the exact base-case and decreasing-ratio checks for

$$26000d^3((7/4)/\alpha)^d < 3/20, \quad d \geq 128.$$

## 7 The full explicit pressure comparison

Let  $N = 2d + 2s$ ,  $0 \leq s \leq d/2 - 1$ , and retain the single positive term

$$P_{d,s} = H_{m,m}[t^N] \tan^{2d} t > \frac{\alpha^d 4^s}{20d},$$

by Lemma 6.2 and the diagonal bound. All frozen coefficients through raw degree  $d$  are positive. The reflected nearest pair in (13) has nonnegative even coefficients, so only its error can contribute a harmful higher frozen coefficient.

Write  $t_R = \arctan R$ . The exact scalar checks give

$$L < 243/200, \quad t_R > 761/2000, \quad \beta = w(243/200) = 11907/20000.$$

For raw degree  $q > d$ , Cauchy's error bound in (13) is  $31000d^2\beta^d R^{-q}$ . Nonnegative tangent coefficients give  $[t^N] \tan^{d+q} t \leq R^{d+q} t_R^{-N}$ . There are at most  $d$  relevant indices, so the complete harmful frozen tail is at most  $31000d^3(R\beta)^d t_R^{-N}$ . This is a finite coefficient sum, not an evaluation of a geometric series at its pole.

Lemma 2.1 bounds the quadratic subtraction by  $C_K(d)g(R)^{2d}t_R^{-N}$ . For  $d \geq 128$ ,  $C_K(d) \leq d^2$ , and  $g(R)^2 < (R\beta)$  because  $RL < w$ . Thus the two errors together are at most

$$31001d^3(R\beta)^d t_R^{-N}. \quad (17)$$

Relative to  $P_{d,s}$ , this is bounded by

$$620020d^4 \rho_w^d, \quad \rho_w = \frac{R\beta}{2\alpha(761/2000)^3} < 1. \quad (18)$$

Indeed the exponential comparison increases with  $\gamma = N/d$  because  $2(761/2000) < 1$ , so using  $\gamma = 3$  enlarges it. Exact rational arithmetic gives

$$620020 \cdot 224^4 \rho_w^{224} < 1/4, \quad \rho_w(225/224)^4 < 1.$$

Consequently (18) is below  $1/4$  for every  $d \geq 224$ .

Finally,  $d \log \cos t$  has coefficient  $-mR_{N/2}/(N/2)$  at even degree  $N$ , where  $0 < R_j \leq 1$ . Its magnitude is at most one. The positive term exceeds four:  $\alpha > 2$  and  $2^{128} > 4 \cdot 20 \cdot 128$  suffice. Equations (3) and (17) now give strict positivity for  $2d \leq N < 3d$ . The positive triangle covers  $d < N < 2d$ . Since  $d = 2m$ , this proves Theorem 1.1 for every  $m \geq 112$ .

## 8 Exact completion of the finite orders

The analytic proof leaves  $2 \leq m \leq 111$ . We give the finite proof specification, including the conversion to the pressure normalization. All rounding below is directed integer rounding; no floating-point approximation is used to decide a sign.

### 8.1 A polynomial gauge and an integer spectral inverse

Let  $D = 2^{d-1}$  and define the atomic Fourier matrix

$$A_{k\ell} = D^{-1} \binom{d}{m+2k-\ell}, \quad 1-m \leq k, \ell \leq m-1,$$

where a binomial coefficient outside  $[0, d]$  is zero. With  $\mathcal{D}(t) = \text{diag}(e^{2ikt})$ , direct expansion gives

$$V_a = (1 + a^2)^m \mathcal{D}(-2t) A \mathcal{D}(t), \quad a = \tan t.$$

Conjugating by  $\mathcal{D}(t)$  therefore gives a similar polynomial matrix

$$M_a = (1 + a^2)^m \mathcal{D}(-t)A, \quad (M_a)_{k\ell} = (1 - ia)^{m+k} (1 + ia)^{m-k} A_{k\ell}.$$

It has exactly the same normalized eigenvalue  $\lambda(a)$  as in (1). Put  $b = ia$ . Then

$$M_a = \sum_{j=0}^d b^j \text{diag}(K_j(k))A, \quad K_0 = 1, \quad K_1 = -2k, \quad (j+1)K_{j+1} = -2kK_j - (d-j+1)K_{j-1}.$$

Each division in this recurrence is exact. Also  $|K_j(k)| \leq \binom{d}{j}$ .

The matrix  $A$  is column-stochastic. Normalize the Fourier eigenvector by the sum of its coordinates, writing  $h(b) = \sum h_n b^n$ ,  $\lambda(b) = \sum \ell_n b^n$ , with  $\sum h_0 = 1$  and  $\sum h_n = 0$  for  $n > 0$ . This gauge normalization changes no eigenvalue. The initial vector is the Eulerian distribution of order  $v = d - 1$ .

Here is a convenient exact inverse. Let  $\mathcal{A}_p(z)$  denote the Eulerian polynomial of order  $p$ , and let  $E_r$  be the coefficient vector of

$$(1 - z)^r \mathcal{A}_{v-r}(z), \quad 0 \leq r \leq v - 1.$$

Define the integer dual matrix

$$S_{ri} = [u^{v-r}] \prod_{h=-i}^{v-1-i} (u + h), \quad 0 \leq i \leq v - 1.$$

Then  $SE = v!I$  and  $AE_r = 2^{-r}E_r$ . For the first identity, the coefficient polynomial of  $P(z)/(1 - z)^{v+1}$  is  $\sum_i p_i \binom{v+k-i}{v}$ ; at  $u = k + 1$ , the Eulerian generating identity sends  $E_r$  to  $u^{v-r}$ . This proves the stated dual transform. The spectral identity also follows from the atomic eigenbasis, and both identities are checked by integer multiplication for every computed  $m$ .

Let  $Qv = v - (\sum v)h_0$  and  $\mathcal{T} = (I - A)^{-1}Q$ . Thus

$$\mathcal{T} = \frac{1}{v!} E \text{diag}\left(0, \frac{2}{2-1}, \dots, \frac{2^{v-1}}{2^{v-1}-1}\right) S.$$

Reflection gives  $h_n(-k) = (-1)^n h_n(k)$ . The even and odd parity blocks have dimensions  $m$  and  $m - 1$ ; the induced full Fourier  $\ell^1$  norm has weights  $1, 2, \dots, 2$  in the even block and weights  $2$  in the odd block. Using the least common multiple of  $2^r - 1$  in each block makes the inverse an integer matrix divided by one integer. Its exact norm  $\alpha_{\pm}$  is the maximum weighted absolute column sum, divided by that column's input weight. The program checks these rational norms without estimation.

## 8.2 Directed enclosure of every response order

Fix the integer grid  $\mathcal{S} = 2^{256}(v!)^5$ . At each order store integers  $u_n, E_n, \hat{\ell}_n, e_n$  satisfying

$$\|\mathcal{S}h_n - u_n\|_1 \leq E_n, \quad |\mathcal{S}\ell_n - \hat{\ell}_n| \leq e_n.$$

The initial vector is exact. Odd scalar coefficients vanish by reflection; all scalar coefficients at  $0 < n < d$  vanish by (1). These structural zeros and their radii are imposed exactly. Every  $u_n$  has exact parity and exact zero normalization for  $n > 0$ .

Put  $B_0 = DA$  and cache  $B_0 u_n$ . The positive-insertion numerator and its error bound are

$$r_n = \sum_{j=1}^{\min(n,d)} \text{diag}(K_j) B_0 u_{n-j}, \quad E_b = \sum_{j=1}^{\min(n,d)} \binom{d}{j} E_{n-j}.$$

The bound follows from  $\|A\|_1 = 1$  and  $|K_j| \leq \binom{d}{j}$ . At a nonstructural scalar order, use

$$\widehat{\ell}_n = \left\lfloor \frac{\sum r_n}{D} \right\rfloor, \quad e_n = E_b + 1.$$

The exact projected recursion is

$$h_n = \mathcal{T} \left( b_n - \sum_{j=1}^{n-1} \ell_j h_{n-j} \right), \quad b_n = \sum_{j=1}^{\min(n,d)} \text{diag}(K_j) A h_{n-j}.$$

The term  $\ell_n h_0$  is killed by  $Q$ . If  $U_h = \|u_h\|_1$ , the scaled right-side error is at most

$$E_{\text{rhs}} = E_b + \left\lceil \frac{1}{\mathcal{S}} \sum_{j=1}^{n-1} (|\widehat{\ell}_j| E_{n-j} + e_j U_{n-j} + e_j E_{n-j}) \right\rceil.$$

Apply the exact integer inverse to the approximate right side

$$\frac{z_n}{D\mathcal{S}}, \quad z_n = \mathcal{S}r_n - D \sum_{j=1}^{n-1} \widehat{\ell}_j u_{n-j},$$

and round each positive-frequency coordinate downward. At even orders set the zero-frequency coordinate to minus twice the sum of those coordinates. The full rounding cost is below  $4(m-1)$  at even orders and  $2(m-1)$  at odd orders. Consequently the valid new radius is

$$E_n = \lceil \alpha_{\pm} E_{\text{rhs}} \rceil + \begin{cases} 4(m-1), & n \text{ even}, \\ 2(m-1), & n \text{ odd}. \end{cases}$$

The two integer inequalities defining each floor are checked. Induction proves the entire enclosure, independently of how large its integers become.

### 8.3 Exact pressure intervals and independent residuals

Return from  $b$  to  $a$ : put  $v_j = (-1)^j \widehat{\ell}_{2j}$  and  $e'_j = e_{2j}$ . Below degree  $3d$  the logarithm is exactly its linear term minus one half its square. Its degree- $2j$  coefficient is enclosed by  $(C_j \pm R_j)/(2\mathcal{S}^2)$ , where

$$C_j = 2\mathcal{S}v_j - \sum_i v_i v_{j-i},$$

$$R_j = 2\mathcal{S}e'_j + \sum_i (|v_i| e'_{j-i} + |v_{j-i}| e'_i + e'_i e'_{j-i}).$$

Only  $i, j-i \geq m$  contribute. Define the nonnegative integers

$$T_{kj} = (2k)! [t^{2k}] \tan^{2j} t, \quad T_{00} = 1.$$

Twice differentiating gives

$$T_{k+1,j} = 2j(2j-1)T_{k,j-1} + 8j^2 T_{kj} + 2j(2j+1)T_{k,j+1}.$$

For  $k \geq 2$ , the cosine term is  $-2mT_{k-1,1}/(2k)!$ . Hence the exact pressure interval is

$$\frac{\sum_j C_j T_{kj} - 4m\mathcal{S}^2 T_{k-1,1} \pm \sum_j R_j T_{kj}}{2\mathcal{S}^2 (2k)!}. \quad (19)$$

Every requested lower numerator must be strictly positive. A nonpositive lower endpoint would be inconclusive and require more precision; it would not certify a negative coefficient.

An independent verifier avoids the producer's transformed solution. Given each stored vector, it recomputes its projected Fourier defect

$$\Delta_n = (I - A)u_n - Q \frac{z_n}{DS}.$$

Fresh radii are propagated with  $E_n^* = \lceil \alpha_{\pm}(E_{\text{rhs}}^* + \|\Delta_n\|_1) \rceil$ , using fresh incoming scalar and vector error radii. Indeed  $\mathcal{T}(I - A)u_n = u_n$  on the exactly normalized subspace. This is an a posteriori proof of the stored vectors that does not rely on their claimed production radii or inverse evaluation. The second verifier also generates tangent by the differential equation  $y' = 1 + y^2$  and ordinary derivative-jet products, independently of the second-derivative recurrence above.

**Proposition 8.1.** *For every integer  $2 \leq m \leq 111$ , every coefficient in Theorem 1.1 is strictly positive.*

*Proof.* The supplied finite certificates execute the preceding integer proof for each of the 110 orders. The degree lists are exactly  $2m, 2m+2, \dots, 6m-2$ , with zero contained at  $2m$  and strictly positive lower endpoints at all 12320 required degrees. There are 37070 nonconstant response orders. Every case succeeds at grid precision 256; no adaptive retry is needed. The programs verify the spectral identities, rounding conditions, normalizations, pressure conversion and exact coverage. The independent residual computation checks every stored state and recomputes the coefficient enclosures. Reproduction instructions and complete per-order hashes accompany the pressure data.  $\square$

Together with the analytic argument for  $m \geq 112$ , this proves Theorem 1.1.

*Editorial note (ProveIt, 2026-10-01).* As filed, the pressure intervals behind Proposition 8.1 are missing: the five interval archives named in `README.md` and `chunks_manifest.json` (`data/interval_mNNN.json` for  $2 \leq m \leq 111$ , about 36 MB) were not delivered, and only their hashes are filed (`finite/production_manifest.json`). So `finite/check_coverage.py` and `finite/check_exact_reference.py` cannot run on the filed files. The finite range is certified again, with separate certificates, by `../Thue_Morse_Integer_Pressure_First_Negative_Data/`, which extends the method of this section to degree  $8m-2$  with freshly propagated error radii: its filed intervals have strictly positive lower endpoints at every even degree strictly between  $2m$  and  $6m$  for  $2 \leq m \leq 128$ , and for  $m = 24$  they enclose all 48 exact values of `finite/exact_reference_m024.json` at degrees 48 to 142 (both checked on filing).

## 9 Sharp endpoint, reproducibility and open questions

At  $m = 2$ , the pressure eigenvalue  $\mu(t)$  near one satisfies

$$\mu^3 - (\cos 2t + 3/4)\mu^2 + (5 \cos 2t/8 + 1/4)\mu - 1/8 = 0.$$

This follows directly by taking the determinant of the three-mode phase matrix. Its root is simple at  $t = 0$ . Exact rational series recursion gives

$$[t^{12}]p_2(t/\pi) = -\frac{35360872}{93555} < 0.$$

Thus the universal positive interval cannot include its first excluded even degree  $6m$ . This does not say that the coefficient at  $6m$  is negative for every  $m$ .

The source bundle includes the complete scalar interval checker, Green and entropy constants, the weighted cutoff, the integer finite producer and independent residual verifier, and the earlier

positive-triangle proof used as an explicit input. Pressure intervals are split into small archives. Full state traces can be regenerated deterministically from the producer; their per-order hashes identify every state without requiring multi-gigabyte attachments. Hashes serve provenance, while the arithmetic replays establish the mathematics. Fully reduced exact calculations at selected orders supply additional independent comparisons.

The stronger raw-coefficient inequality  $[a^{d+2s}]F + 2[a^{2s}]FG \geq 0$  for  $0 \leq s < m$  remains open. Finite-iterate positivity, termwise return-depth positivity and positivity in the atomic-coordinate cone have exact counterexamples and are not proof premises. The location of the first negative coefficient beyond the proved interval, and any proportional asymptotic law for that location, are separate questions.

*Editorial note (ProveIt, 2026-10-01).* Both are answered by later packages of this series. The first negative degree  $N_m$  above  $2m$  exists for every  $m \geq 2$  (`../Thue_Morse_Integer_Pressure_Sign_Changes/`), so  $N_m \geq 6m$  by Theorem 1.1, and  $N_m = \gamma m - \Lambda^{-1} \log \log(2m) + O(1)$  with  $\gamma \in (6.662966, 6.662968)$  (`../Thue_Morse_Integer_Pressure_First_Negative/`). The dataset `../Thue_Morse_Integer_Pressure_First_Negative_Data/` certifies  $N_m$  for  $2 \leq m \leq 128$ :  $N_m = 6m$  for  $m = 2, 3, 4$ , while for  $5 \leq m \leq 128$  the coefficient at degree  $6m$  is positive.

*Editorial note (ProveIt, 2026-10-01).* This note is the seventh of thirteen packages written in one series and filed together on 2026-10-01 beside the repository manuscript `../Thue_Morse_Integer_Pressure/` in the Fabius drafts tree (batch 72 of the repository's `docs/incoming/`; unreviewed). In logical order they are the positivity series `../Thue_Morse_Integer_Pressure_First_Positive/`, `../Thue_Morse_Integer_Pressure_Higher_Positive/`, `../Thue_Morse_Integer_Pressure_Positive_Triangle/`, `../Thue_Morse_Integer_Pressure_Feedback_Boundary/`, `../Thue_Morse_Integer_Pressure_Beyond_Boundary/`, `../Thue_Morse_Integer_Pressure_Linear_Region/` and `../Thue_Morse_Integer_Pressure_Full_Range/`, and the sign series, which imports the full-range theorem: `../Thue_Morse_Integer_Pressure_Sign_Changes/`, `../Thue_Morse_Integer_Pressure_Sign_Densities/`, `../Thue_Morse_Integer_Pressure_Negative_Bound/`, `../Thue_Morse_Integer_Pressure_First_Negative/` and `../Thue_Morse_Integer_Pressure_Cluster_Asymptotics/`, with the dataset `../Thue_Morse_Integer_Pressure_First_Negative_Data/` (no article). An earlier version of this article, *The Full Pressure Positivity Range for Large Integer Orders* (orders  $m \geq 4096$ ), delivered in the same batch, is superseded by it and was not filed. Here [1] is `../Thue_Morse_Integer_Pressure/` (the version cited differs from the filed manuscript only by dated editorial changes, none of them mathematical), [2] is `../Thue_Morse_Integer_Pressure_Positive_Triangle/` (the copy bundled under `inputs/` was not filed), and [3] is `../Thue_Morse_Integer_Pressure_Feedback_Boundary/`. The superseded version, which this article does not cite, claimed the theorem for  $m \geq 4096$  only.

## References

- [1] *Integer Pressure and a Missing Taylor Coefficient*. ProveIt research manuscript, inspected commit 6a98f89bac85cc6b4ba5d5c3b63996037d9824e1, blob 1973fd17245864bb8a83e1b1d90416ac032ba7c1. Versioned repository source.
- [2] *The Full Positive Triangle in Integer Thue Morse Pressure*. Research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026. The dyadic tangent comparison and positivity for  $1 \leq r < m$ . Included unchanged in the source bundle.
- [3] *Positivity at the First Feedback Boundary in Thue Morse Pressure*. Separate research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026. The earlier all-order degree- $4m$  result; the large-order diagonal lower bound used here is proved anew.